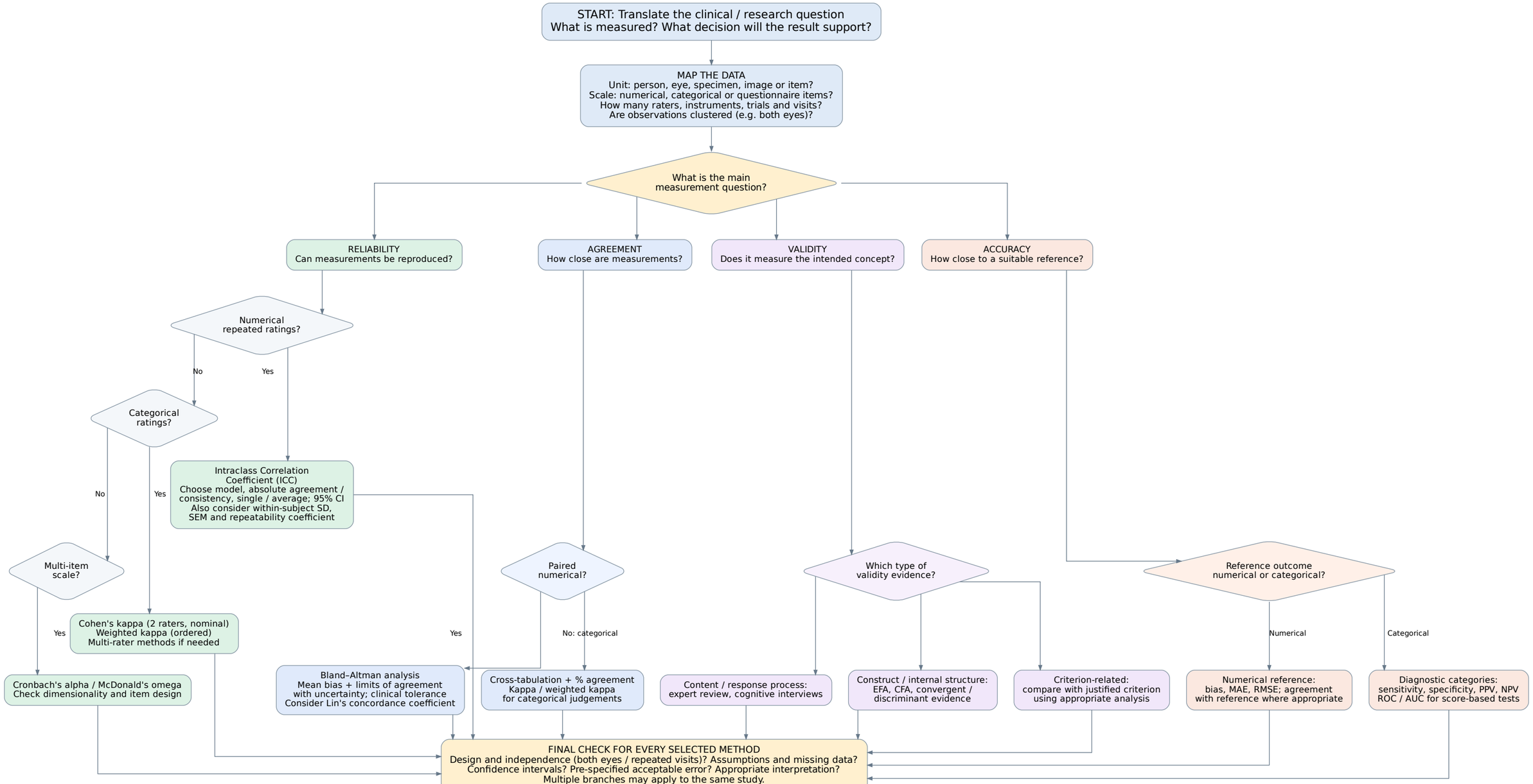
START: Translate the clinical / research question
What is measured? What decision will the result support?
MAP THE DATA
Unit: person, eye, specimen, image or item?
Scale: numerical, categorical or questionnaire items?
How many raters, instruments, trials and visits?
Are observations clustered (e.g. both eyes)?
What is the main measurement question?
RELIABILITY
Can measurements be reproduced?
AGREEMENT
How close are measurements?
VALIDITY
Does it measure the intended concept?
ACCURACY
How close to a suitable reference?
Numerical repeated ratings?
No
Yes
Categorical ratings?
No
Yes
Intraclass Correlation Coefficient (ICC)
Choose model, absolute agreement / consistency, single / average; 95% CI
Also consider within-subject SD, SEM and repeatability coefficient
Multi-item scale?
Yes
Cohen's kappa (2 raters, nominal)
Weighted kappa (ordered)
Multi-rater methods if needed
Cronbach's alpha / McDonald's omega
Check dimensionality and item design
Paired numerical?
Yes
No: categorical
Bland–Altman analysis
Mean bias + limits of agreement
with uncertainty; clinical tolerance
Consider Lin's concordance coefficient
Cross-tabulation + % agreement
Kappa / weighted kappa
for categorical judgements
Which type of validity evidence?
Content / response process:
expert review, cognitive interviews
Construct / internal structure:
EFA, CFA, convergent / discriminant evidence
Criterion-related:
compare with justified criterion
using appropriate analysis
Reference outcome numerical or categorical?
Numerical
Categorical
Numerical reference:
bias, MAE, RMSE; agreement
with reference where appropriate
Diagnostic categories:
sensitivity, specificity, PPV, NPV
ROC / AUC for score-based tests
FINAL CHECK FOR EVERY SELECTED METHOD
Design and independence (both eyes / repeated visits)? Assumptions and missing data?
Confidence intervals? Pre-specified acceptable error? Appropriate interpretation?
Multiple branches may apply to the same study.